

Ellen Poulsen

 Ellen Poulsen |  ellenpoulsen5@gmail.com |  +1 (541) 654-2129

EDUCATION

2025 — 2029 **B.S. Aerospace Engineering Degree Candidate** GPA: 3.66/4.0
California Polytechnic State University, San Luis Obispo

WORK EXPERIENCE

Test Engineering Intern — Empirical Systems Aerospace *June 2026 — Aug. 2026*
San Luis Obispo, CA

- Developed servo testing procedures and software to analyze frequency response, working in parallel with mechanical design team on servo test stand to accurately measure angular velocity and torque.
- Hosted preliminary design review demonstrating technical feasibility and supporting consideration of the system for R&D and production applications.
- Designed Python based data acquisition software to measure phase delay and bandwidth, with focus on graphical data representation and user interface for broad usage requirements.
- Operated with both CAN bus and PWM communication protocols for use across different vehicles.
- Calibrated data acquisition timing across multiple test devices to capture precise telemetry during high-speed, high-load actuation while preventing data loss caused by communication bandwidth limitations.
- Managed project version control and code collaboration using Python, VS Code, and GitLab, alongside thorough documentation of product development for project handoff and future development.

Observatory Assistant — Cal Poly Physics Department *May 2026 — Present*

- Operating 14-inch Meade LX600 ACF Telescope for public viewing events.
- Leading educational observing sessions for student groups, presenting astrophysics concepts while tracking night sky targets.

PROJECTS

Cal Poly Space Systems — Liquid Propulsion Engineer

- Development of liquid rocket engines including co-leading work on current regenerative engine project: developing testing dates and metrics, task delegation, and goals for future vehicle integration.
- Performing high pressure fluids testing safely through continuously updated procedures, familiarization with modes of failure including back flow prevention with pressure ladder margins, and engine hardstart mitigation through propellant lead sequencing and ignition timing.
- Improving and maintaining test stand by ensuring proper operation and reading from pressure transducers, maintaining oxygen compatible hardware, and testing proper operation of pneumatic and electrical systems.

LEADERSHIP EXPERIENCE

Cal Poly Space Systems Lead — SILLE (Student Introductory Liquids Learning Engine)

- Teaching students how to design and manufacture their own small liquid rocket engine, present and receive feedback from design reviews, and conduct safe testing on their bipropellant engines.
- Organization of over 40 students into teams, teaching basics of liquids rocketry, foundational understanding of isentropic equations and associated variables, machining, and safe testing practices.
- Focus on member retention and transfer of knowledge through dedicated mentorship, ensuring students feel supported and encouraging their continued involvement to promote longevity and growth of the liquids team.
- Public speaking and academic presentation to large audiences alongside proficiency in group organization and communication utilizing Microsoft Teams, PowerPoint, Forms, and Excel.

SKILLS

Software: familiar with Python, MATLAB, LaTeX; SolidWorks, GitHub, VSCode, Microsoft Office Suite

Manufacturing: Manual Mill and Lathe, Welding (MIG and TIG)

Languages: Spanish (Working Proficiency)